🤌 XAI: What is Your Data Worth? Equitable Valuation of Data

FAMAF - UNC

First Semester 2026

🧉 Disclaimer ------

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

What is Your Data Worth? Equitable Valuation of Data

Amirata Ghorbani, James Zou — ICML 2019

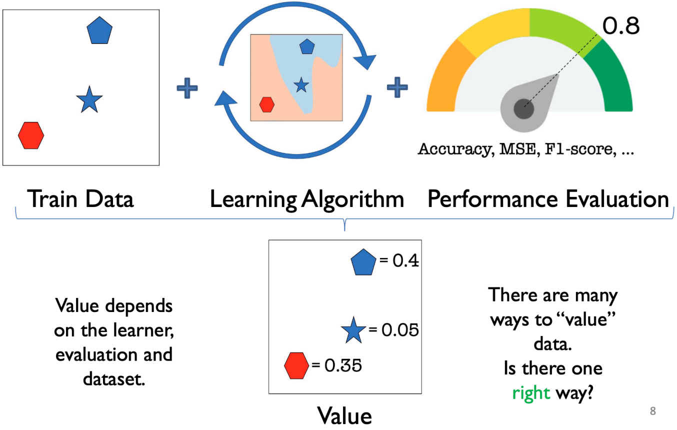
(Ghorbani and Zou 2019)

🧉 The Data Valuation Problem ------

If data is fuel, we need a principled way to measure its value. Stakeholders have different priorities:

- **ML Engineers:** Assess heterogeneous sources and data quality.
- **Data Vendors:** Determine fair pricing for buying and selling data.
- **Individuals/Data Producers:** Understand compensation or credit for contributed data.

🧉 Ingredients of Data Value ------



Value depends on the learner, evaluation and dataset.

There are many ways to "value" data.
Is there one right way?

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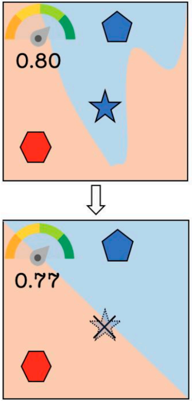
🧉 Leave One Out Method ------

<u>Example</u>: value (★) = 0.80 – 0.77 = 0.03



🧉 Desirable Properties of Valuation ------

A fair valuation method must satisfy three fundamental axioms from cooperative game theory:

1. **Null Element:** If adding a point to *any* subset of training data never changes the model's performance, its value is exactly 0.
2. **Symmetry:** If adding point A or point B to any subset always results in the exact same performance change, they must receive the same value $(Value(A) = Value(B))$.
3. **Linearity:** If the performance metric is a sum of performances on individual tasks, the data value must reflect the sum of values for those individual tasks.

The Data Shapley Value ------

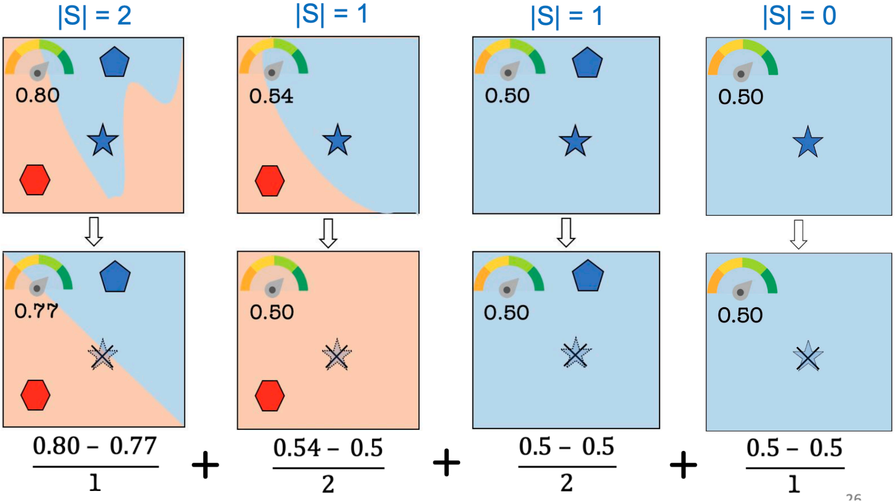
The only data value mathematically guaranteed to satisfy these three properties is based on marginal contributions across all possible subset sizes.

$$Value(data\ k) = \sum_{S \subseteq D\ \{k\}} \frac{performance(S \cup \{k\}) - performance(S)}{\binom{n-1}{|S|}}$$

- Evaluates the expected contribution to all possible sizes of train data samples.
- Directly applies Lloyd Shapley's cooperative game theory to individual ML data points.

🧉 The Data Shapley Value Example ------

Example: value (★) = 0.05



🧉 Efficient Approximation: TMC-Shapley ------

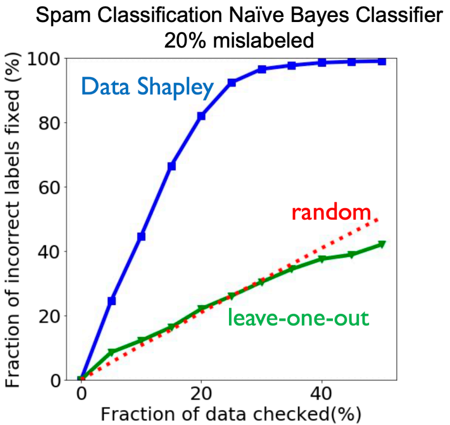
Exact calculation is computationally infeasible for realistic datasets because it grows exponentially.

Truncated Monte-Carlo (TMC) Approximation:

1. Sample a random permutation of the training data.
2. Add one point at a time based on the sampled permutation.
3. Re-train the model to capture the marginal increase in performance.
4. **Truncate** the calculation when performance saturates to save computational resources.

🧉 Application 1: Identifying Low-Quality Data ------

Result: Ordering points from most negative Shapley value to positive allows for rapid dataset cleaning. Examining just ∼30% of the data uncovers nearly all mislabeled examples.



🧉 Application 2: Identifying Essential Data ------

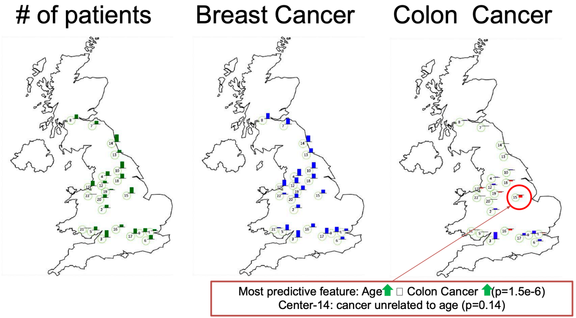
Case Study: UK Biobank

- Dataset spanning 500,000 individuals across 22 centers in the UK.
- Centers were evaluated as singular "data sources" for predicting Breast and Colon Cancer.

🧉 Application 2: Identifying Essential Data ------

Insights from Shapley:

- For Colon Cancer, a massive center received a *negative* Shapley value.
- Investigation revealed the model heavily relied on *Age* as a predictive feature, but the specific anomalous center exhibited colon cancer rates entirely independent of age.



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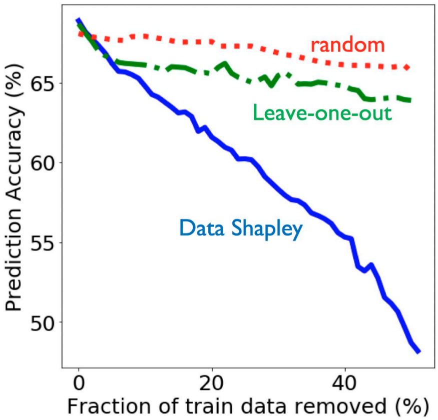
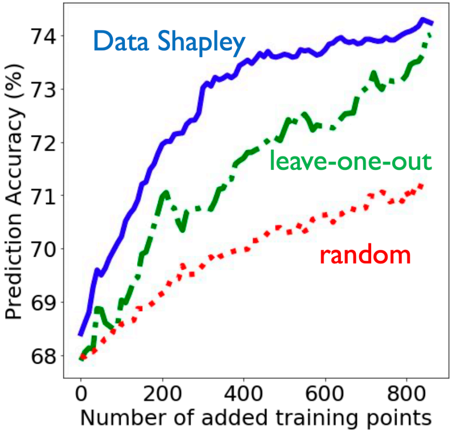




🧉 Application 3: Domain Adaptation ------

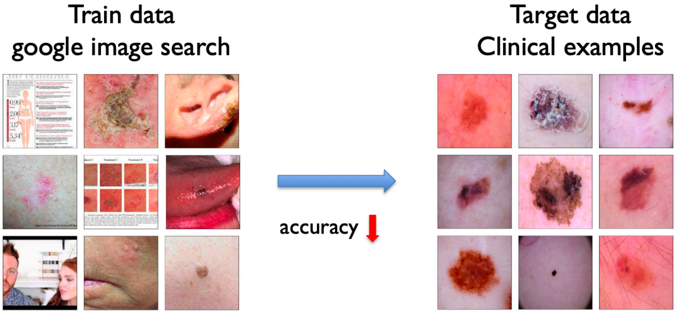
What if the training data differs in quality, distribution, or class balance from the test environment?

Shapley Adaptation Strategy:

1. Compute Shapley values relative to the target test distribution.
2. Remove data with negative values.
3. Reweight the remaining training data based on relative positive weights.

🧉 Domain Adaptation Results (Skin Lesion) ------

Training on noisy Google Image Search data to test on the clean, clinical HAM10000 dataset. Applying Shapley weights improved classification by $\approx 25\%$.



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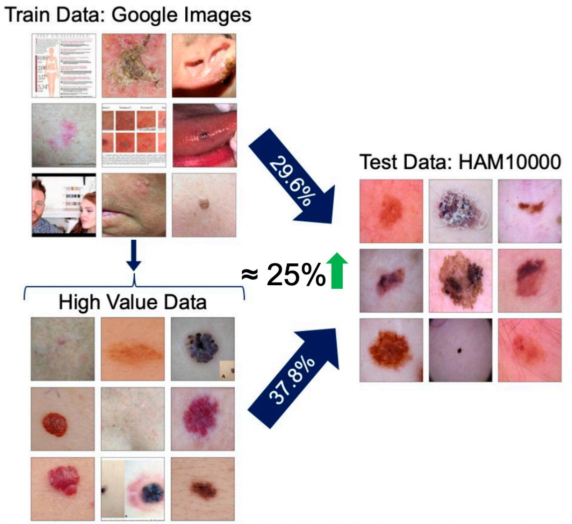
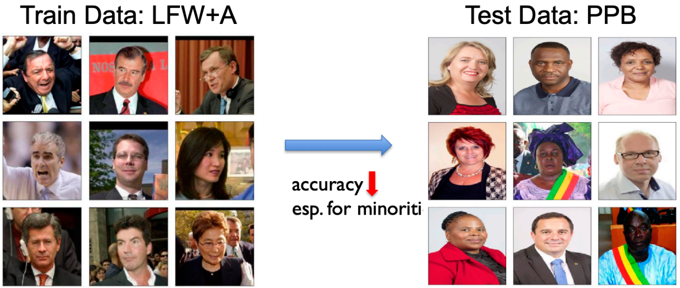


🧉 Domain Adaptation Results (Gender Detection) ------

Models trained on LFW+A (heavily biased toward white males) fail on balanced datasets like PPB. Shapley values easily identify out-of-distribution elements; adaptation increases accuracy on underrepresented minorities by $\approx 7\%$.



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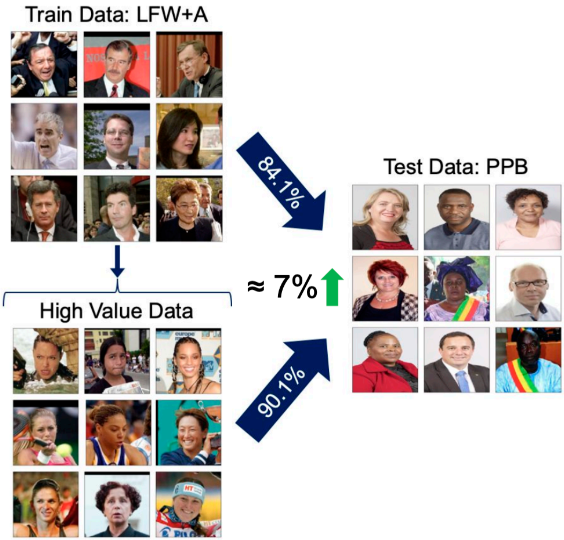


🧉 Summary ------

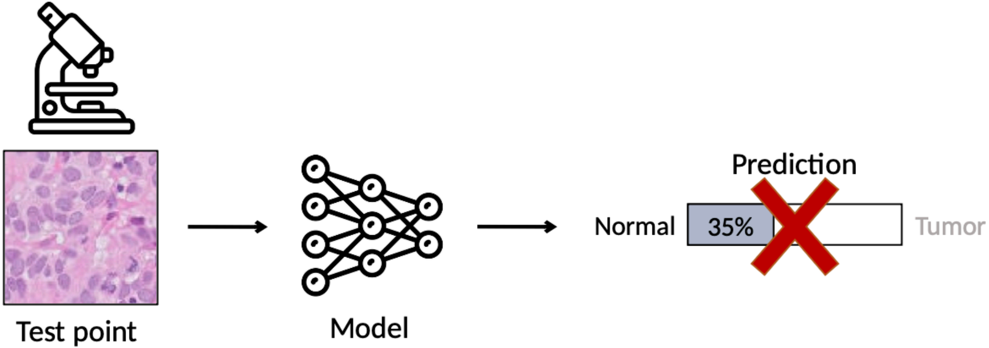
- Data Shapley provides an equitable, axiom-backed framework to quantify the value of individual ML data points.
- It moves beyond LOO limitations by measuring expected contributions over all possible subsets.
- Applications span data cleaning, debugging, valuation pricing, and addressing distributional shifts.

Understanding black-box predictions via influence functions

Pang Wei Koh, Percy Liang — ICML 2017

(Koh and Liang 2017)

🧉 Dataset Debugging ------



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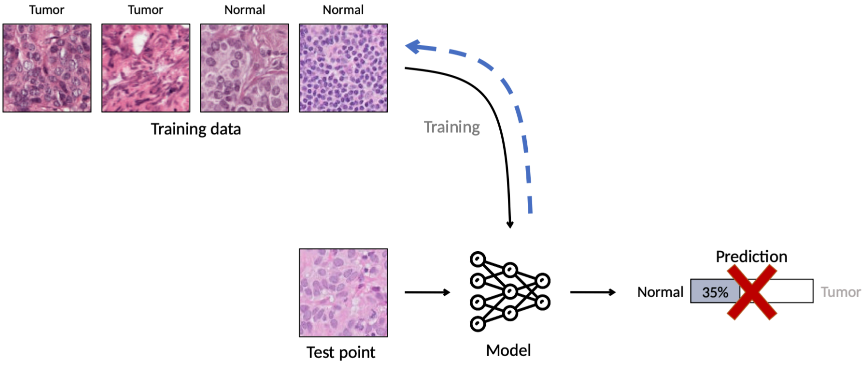

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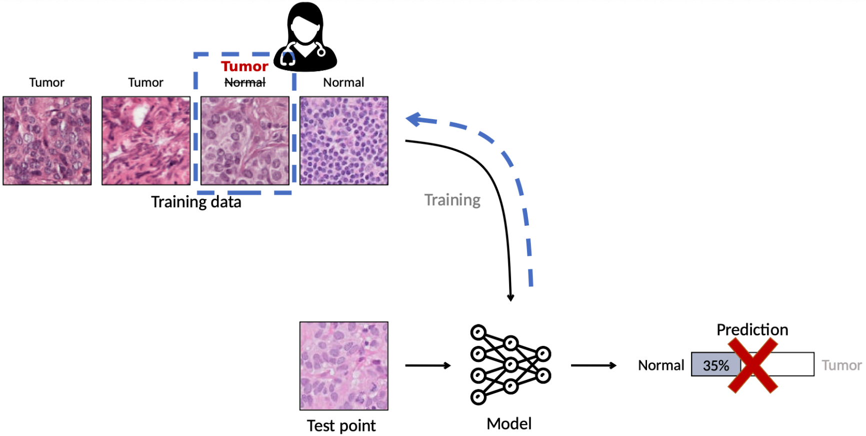

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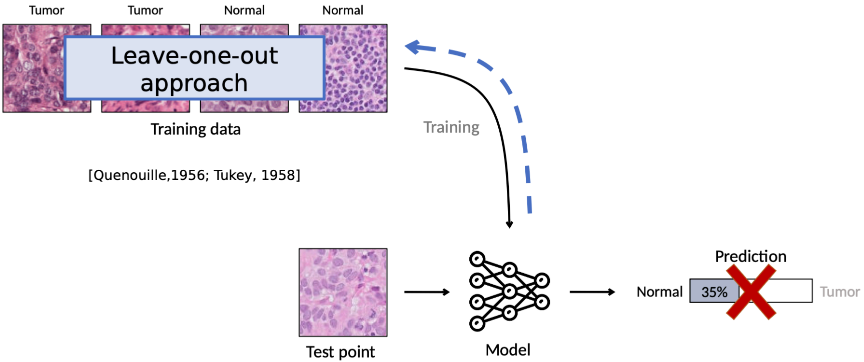
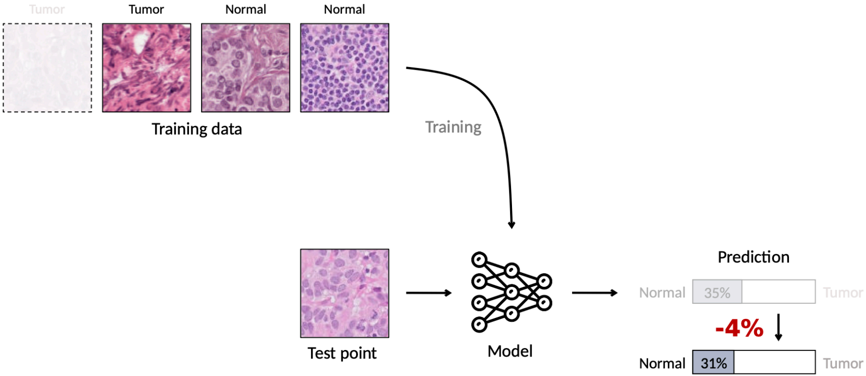


[Quenouille,1956; Tukey, 1958]

Dataset Debugging ------



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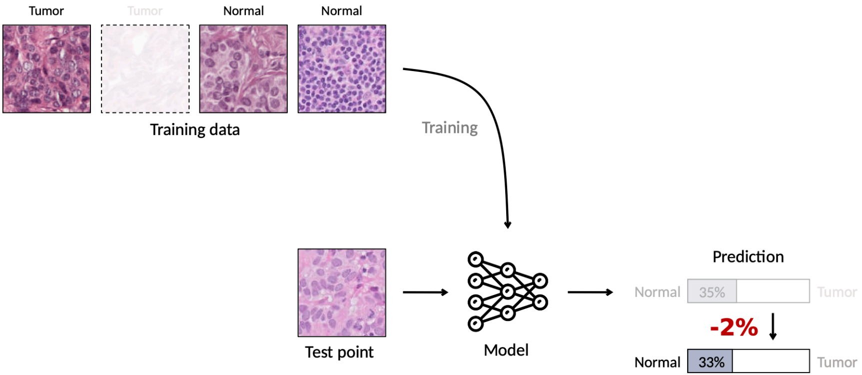

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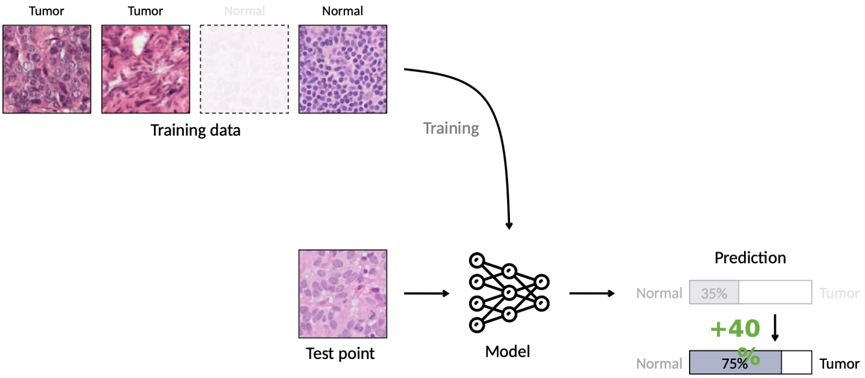

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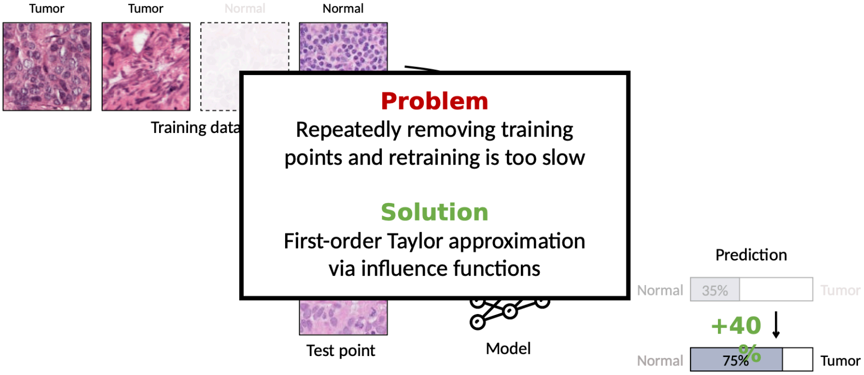

Problem

Repeatedly removing training points and retraining is too slow

Solution

First-order Taylor approximation via influence functions

🧉 The Influence Function Approximation ------

$$\overbrace{\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right)}^{\text{Change in loss on after removing}} \approx \nabla_{\theta} \overbrace{\underbrace{\ell\left(z_{test}, \hat{\theta}\right)^{T}}_{\text{Gradient of loss on}}}^{\text{Model's representation of}} \overset{\text{Effect of other training points}}{\underset{\text{Inverse of the Hessian}}{H_{\hat{\theta}}^{-1}}} \overbrace{\underbrace{\nabla_{\theta} \ell\left(z_{train}, \hat{\theta}\right)}_{\text{Gradient of loss on}}}^{\text{Model's representation of}}$$

Doesn't require retraining!

🧉 Interpretation: Model Representations ------

$$\ell(z_{\text{test}}, \hat{\theta}_{-z_{\text{train}}}) - \ell(z_{\text{test}}, \hat{\theta}) \approx \nabla_\theta \ell(z_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell(z_{\text{train}}, \hat{\theta})$$

Gradient $=$ model's representation

$\nabla_\theta \ell(z_{\text{test}}, \hat{\theta})$ encodes how the model "sees" z_{test}

$H_{\hat{\theta}}^{-1}$ $=$ effect of other training points

Accounts for the curvature induced by the full training set

High influence $\Rightarrow$ test and train have **similar model representations**

🧉 Removing Single Points: Validation ------

Logistic regression (MNIST): each point = removing one training example. Influence function estimate tracks actual LOO change closely.

🧉 Impact ------

Robustness

- Training set biases [Ren et al., 2018]
- Inference reliability [Broderick et al., 2021]
- Cross-validation [Stephenson et al., 2020]
- Memorization [Feldman, 2019]

Applications

- Data distillation [Wang et al., 2020]
- Data valuation [Jia et al., 2019]
- Active learning [Gudovskiy et al., 2020]
- Data debugging [Guo et al., 2021]

Fairness & Security

- Algorithmic bias [Verma et al., 2021]
- Data labor [Arrieta-Ibarra, 2018]
- Privacy [Shokri et al., 2021]
- Data poisoning [Chen et al., 2017]

🧉 Limitations ------

Key assumptions required

- What assumptions on models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network?

$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} L(z_i, \theta) + \epsilon L(z, \theta)$$

- **Non-convexity:** derivation assumes a convex loss landscape
- **Non-convergence:** derivation assumes the optimizer has converged

"If Influence Functions are the Answer, Then What is the Question?"

Bae, Ng, Lo, Ghassemi & Grosse

🧉 Summary ------

- **Link** model behavior to training data
- **Efficient** to calculate — no retraining needed
- **Many applications:** debugging, fairness, security, privacy
- **Limitations** when applying to complex models
  - Non-convexity
  - Non-convergence

Thank You!

🧉 References ------ |

Ghorbani, Amirata, and James Zou. 2019. "Data Shapley: Equitable Valuation of Data for Machine Learning." In *Proceedings of the 36th International Conference on Machine Learning*, edited by Kamalika Chaudhuri and Ruslan Salakhutdinov, vol. 97. Proceedings of Machine Learning Research. PMLR. https://proceedings.mlr.press/v97/ghorbani19c.html.

Koh, Pang Wei, and Percy Liang. 2017. "Understanding Black-Box Predictions via Influence Functions." *International Conference on Machine Learning*, 1885–94.